

AI Usage Documentation

Network Telemetry & Real-Time Network Performance Monitoring System

1. How I Used AI During the Project

I used ChatGPT and Claude mainly as technical assistants while developing and debugging the project. I did not use AI as a replacement for running the project. I used it when I was stuck, when I needed to understand an error, when I wanted to compare implementation options, or when I needed help checking whether a change would affect the rest of the system. The actual commands were run in my own environment, and I checked the output before accepting a change.

2. Prompts I Used During Development

The prompts below represent the kind of prompts I actually used while working through the project. They cover the project from initial understanding through implementation, testing, monitoring and the later GNS3/VyOS integration.

Project area	Prompt I used	How it helped
Understanding the existing project	"I have this Network Telemetry project already started. Go through the repository structure and explain what each major component is doing, especially the collector, gRPC layer, FastAPI backend, database and Grafana. Tell me what I should understand before changing anything."	Helped me understand the existing system before modifying it.
Choosing the architecture	"For this project I need a real-time network telemetry pipeline. Based on the existing code, how should SNMP/ICMP collection, gRPC, FastAPI, PostgreSQL and Grafana fit together? Point out what should stay separate and what should communicate with each other."	Helped me settle the overall data flow and component responsibilities.
Collector implementation	"Inspect the collector logic and help me add/verify SNMP metrics for interface traffic, utilization, packet counts, errors and device reachability. Keep the existing metric structure and don't break the current simulator."	Helped with the collector logic while preserving the existing design.
SNMP/OID understanding	"I'm using pysnmp and IF-MIB. Explain what ifName, ifHCInOctets, ifHCOutOctets, ifInErrors, ifOutErrors, ifHighSpeed and ifOperStatus are giving me, and how I can use them to calculate the telemetry metrics in this project."	Helped me understand the actual SNMP data being collected and how it maps to metrics.
gRPC integration	"Check how the collector and backend communicate through protobuf/gRPC in this project. If I change the telemetry data being sent, what files or message definitions do I need to check so the collector and backend stay compatible?"	Helped me trace data through the collector-to-backend boundary.
PostgreSQL debugging	"PostgreSQL is running in Docker but the backend/collector is not connecting correctly. Here is the error and my compose configuration. Find the likely mismatch and tell me what I should check first without deleting my existing data."	Helped narrow down the database/container configuration problem.
Docker troubleshooting	"My Docker Compose services are up/down like this: [paste docker compose ps and logs]. Explain which service is actually failing and why. Give me the smallest fix and one command I can use to verify it."	Helped me debug Docker services without changing unrelated components.
Grafana dashboard	"My telemetry is already in PostgreSQL. I want Grafana to show RX bandwidth, TX bandwidth and utilization with device and interface filters. Based on the table structure, write the SQL query and explain the variable interpolation I need."	Helped create and correct the Grafana queries and filters.
Testing	"I need to demonstrate that the telemetry system actually works, not just that the services start. Give me practical test cases for normal telemetry, threshold alerts, recovery, unreachable devices, database records and Grafana results. I will run the tests myself."	Helped me make a realistic test checklist.
Interpreting real output	"Here is the actual output from my terminal/API/Grafana. Tell me what it proves, what it does not prove, and whether I should make another test before calling this part complete."	Helped me avoid claiming results that were not actually verified.
GNS3/VyOS planning	"I already have the telemetry application working. I now want to use a GNS3 topology with VyOS	Helped me plan the move from simulated

	routers as the real SNMP source. How can I add the virtual routers without replacing the existing telemetry pipeline?"	sources toward a real virtual network source.
GNS3/VirtualBox troubleshooting	"My VyOS router is running inside a GNS3 VM in VirtualBox. Windows can reach the GNS3 VM management IP but cannot ping the VyOS router IP. Explain what network path is missing and what information I should check before changing anything."	Helped identify that the GNS3 VM management network and the isolated router topology were not automatically the same network.
VyOS interface issue	"After cloning the VyOS router, the interface names changed. Here is `show interfaces`. Which interface is actually up, and how should I move the existing saved management IP to the interface that is physically connected? Don't make me reinstall the router."	Helped interpret the cloned-router interface mismatch.
Final integration check	"I want the final architecture to be GNS3/VyOS -> SNMP + ICMP -> Python Collector -> gRPC -> FastAPI -> PostgreSQL -> Grafana. Check the existing project and tell me the minimum configuration changes needed to add the VyOS device while keeping the rest of the system intact."	Helped connect the virtual network to the existing telemetry architecture.
Understanding SNMP in my project	I understand that the collector is using SNMP, but I want to understand what is actually happening between the router and collector. Explain SNMPv2c, the community string, polling, OIDs and the IF-MIB values we are collecting. Then connect each value to the bandwidth, utilization, packet/error and interface metrics in my project.	Helped me understand the SNMP flow and why the specific OIDs were needed instead of treating SNMP as a black box.

3. How I Worked With the AI Suggestions

My workflow was not to copy an answer and assume it was correct. I normally gave AI the actual error, configuration or output I was seeing, asked what it meant, and then applied the relevant change myself. This was especially important for environment-level problems because the AI could not directly see my Windows machine, VirtualBox configuration, GNS3 VM, running containers or router state.

There were also cases where an AI suggestion did not work as expected. For example, while working with GNS3/VyOS I had QEMU, console and networking issues. I had to compare the suggested fix with the actual GNS3 output and change direction. Similarly, database, SNMP and Python environment issues were resolved by checking the real command output rather than assuming that the first suggestion was correct.

4. What AI Contributed vs. What I Did

Stage	AI contribution	My contribution
Planning	Discussed architecture, component roles and possible approaches.	Defined the project requirements and chose what to implement.
Implementation	Suggested code, SQL, YAML, Docker and configuration changes.	Applied, modified and integrated the changes in the actual project.
Environment setup	Explained setup issues and possible causes.	Configured Windows, Docker, VirtualBox and GNS3 and ran the commands.
Debugging	Interpreted errors and suggested what to inspect next.	Collected the actual logs/output, tested fixes and decided what worked.
Testing	Suggested test scenarios and evidence to collect.	Executed the tests and recorded the real results.
Evaluation	Helped interpret API, database and dashboard output.	Verified that the observed result matched the project requirement.
Documentation	Helped organize and improve wording.	Finalized the content using the work and results from my project.